SOLANA

DEEP-DIVE RESEARCH SERIES

FILE 1 OF 3

ORIGINS & TECHNOLOGY

Founding Story · Team Biographies · Blockchain Trilemma
Proof of History · 8 Core Innovations · Transaction Lifecycle

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01 — EXECUTIVE SUMMARY

Solana is a high-performance, open-source blockchain platform designed to enable decentralised applications and crypto-currencies at a scale previously considered impossible without sacrificing decentralisation or security. Since its mainnet beta launch in March 2020, it has grown into one of the top blockchain ecosystems globally, processing hundreds of millions of transactions and hosting thousands of decentralised applications across DeFi, NFTs, gaming, and payments.

At the heart of Solana's architecture is a novel cryptographic clock called **Proof of History (PoH)**, invented by founder Anatoly Yakovenko. Unlike traditional blockchains that must spend precious consensus time agreeing on *when* events occurred, Proof of History encodes the passage of time directly into the chain — allowing validators to process, verify, and agree on the order of transactions in parallel rather than sequentially. This single innovation unlocks a theoretical throughput of 65,000+ transactions per second (TPS), with sub-second finality.

This report — the first of three — covers everything from the founding story and the team behind Solana, to a comprehensive breakdown of the eight core innovations that make the network uniquely fast, secure, and scalable.

METRIC	VALUE
Founded	2017 (whitepaper) · 2018 (Solana Labs)
Mainnet Launch	March 2020
Peak TPS (Theoretical)	65,000+
Average Block Time	~400 milliseconds
Total Raised	$335 million+
Native Token	SOL
Consensus Mechanism	Proof of History + Tower BFT (PoS)
Programming Language	Rust, C, C++

02 — FOUNDING STORY & HISTORY

The Problem Anatoly Couldn't Ignore

The story of Solana begins not in a Silicon Valley garage, but inside the mind of a seasoned systems engineer frustrated by a fundamental limitation of Bitcoin and Ethereum. **Anatoly Yakovenko**, a Ukrainian-born software engineer who had spent over a decade at Qualcomm working on operating systems and distributed systems, was deeply familiar with how to make hardware communicate at extraordinary speed.

In late 2017, while reading about Bitcoin and Ethereum's throughput bottlenecks, Yakovenko had an insight that would change blockchain architecture. He noticed that neither Bitcoin nor Ethereum had a reliable, trustless way to tell *time*. Every node on those networks relied on its own local clock, and since nodes couldn't trust each other's clocks (an adversarial environment), every block had to include timestamps that were loosely enforced. This meant the entire network had to slow down and agree not just on *what* happened, but *when* — an enormous waste of time and throughput.

The Core Insight

"What if we could create a trustless, decentralised clock that could prove the passage of time without needing every node to agree on a timestamp?"

— Anatoly Yakovenko, late 2017

This question led directly to the invention of Proof of History — a cryptographic mechanism that encodes time into the blockchain itself, eliminating one of the biggest bottlenecks in distributed consensus.

The Whitepaper — November 2017

Yakovenko published the original Solana whitepaper in November 2017, introducing Proof of History to the world. The paper described a method of creating a historical record that proved that an event had occurred at a specific moment in time, using a verifiable delay function (VDF) — specifically, a sequential SHA-256 hash chain. Each hash in the chain takes a calculable amount of time to compute and its output becomes the input of the next, creating an immutable, verifiable sequence of events with embedded time.

The whitepaper circulated quickly among cryptography and distributed systems circles. Greg Fitzgerald, a former Qualcomm colleague, read it and immediately reached out to Yakovenko. He was so convinced by the concept that he left his job within weeks and became the first co-founder to join. Fitzgerald wrote the first implementation of the protocol in the Rust programming language — a choice that would define Solana's technical DNA.

Founding Solana Labs — 2018

In January 2018, Anatoly Yakovenko officially founded Solana Labs in San Francisco. The founding team assembled quickly: Greg Fitzgerald (CTO), Raj Gokal (COO), Stephen Akridge, Eric Williams, and several other engineers who had backgrounds at Qualcomm, Apple, Intel, Google, and Dropbox. The company was initially named **Loom**, but had to rename itself when a competing Ethereum project took that name. It was renamed **Solana** — inspired by Solana Beach, a coastal town north of San Diego where Yakovenko and several team members had previously lived and surfed.

Early Fundraising

ROUND	DATE	AMOUNT	KEY INVESTORS
Seed	Q2 2018	$3.17M	Multicoin Capital, Foundation Capital
Series A	July 2019	$20M	Multicoin Capital, Distributed Global
Strategic	Q1 2020	$1.76M	Various angels
Series A Ext.	July 2020	$1.76M	Andreessen Horowitz (a16z)
Series B	June 2021	$314.15M	a16z, Polychain Capital, Coatue

Solana's Series B in June 2021 — raising $314.15 million — was one of the largest single fundraising rounds in blockchain history at the time. The peculiar number ($314.15M ≈ 100π million) was a deliberate nod to the mathematical constant, a signature of the engineering culture at the company.

Mainnet Beta — March 2020

After nearly two years of development and four testnet phases (Tour de SOL), Solana launched its **Mainnet Beta** in March 2020 — the same month the COVID-19 pandemic began reshaping the world. Despite the turbulent backdrop, Solana immediately demonstrated its performance promises: processing thousands of transactions per second with block times around 400 milliseconds. The Solana Foundation, a separate non-profit entity, was also established in 2020 to support the long-term development of the ecosystem.

03 — FOUNDING TEAM BIOGRAPHIES

Solana was built by a team of world-class engineers, many of whom had spent years working on complex distributed systems at some of the world's most demanding technology companies. Their deep hardware and systems background is what made Solana's approach to blockchain fundamentally different.

Anatoly Yakovenko — CEO & Co-Founder

Born in Dnipro, Ukraine, Anatoly Yakovenko studied Computer Science at the University of Illinois Urbana-Champaign before moving to San Diego. He spent 12+ years at Qualcomm, rising to Principal Engineer, working on operating systems for mobile chipsets — systems requiring microsecond-level timing precision and massive parallel processing. Before founding Solana, he also worked at Dropbox as a software engineer. It was his obsession with clock synchronisation and distributed systems timing that led to the invention of Proof of History. Yakovenko is widely regarded as one of the most technically rigorous founders in crypto, known for deep-dive technical threads explaining blockchain design with exceptional clarity.

Greg Fitzgerald — CTO & Co-Founder

Greg Fitzgerald is the engineer who transformed Yakovenko's theoretical Proof of History whitepaper into working code. A fellow Qualcomm veteran, Fitzgerald is the primary author of the Solana protocol's original Rust implementation. He is known within the Solana developer community as the person who wrote the first validator client and established the architectural patterns that still define Solana's codebase today. Fitzgerald has a particular passion for programming language design and compiler infrastructure — skills that align perfectly with Solana's use of Rust, a language designed for memory safety and high performance.

Raj Gokal — COO & Co-Founder

Raj Gokal brings the product, business, and venture perspective to Solana's founding team. Before Solana, he was a product manager and investor, with experience at Omada Health and General Catalyst (a major venture capital firm). Gokal is responsible for Solana's go-to-market strategy, ecosystem development, and investor relations. He has been instrumental in building Solana's developer community and forging partnerships with major protocols. Gokal is often the face of Solana at major industry conferences and is known for his ability to communicate the network's technical advantages to non-technical audiences.

Stephen Akridge — Co-Founder

Stephen Akridge is one of the quieter but technically critical co-founders of Solana. A former Qualcomm engineer, Akridge's key contribution was the invention of the **Seagull** GPU-based signature verification system. His insight that graphics processing units could be repurposed to verify cryptographic transaction signatures in massive parallel batches gave Solana a significant throughput advantage over CPU-only approaches. This contribution directly feeds into Solana's ability to process tens of thousands of transactions per second. Akridge later co-founded Cypher, a Solana-based decentralised exchange.

Eric Williams

Chief Scientist & Co-Founder

Eric Williams serves as Solana's Chief Scientist and is responsible for the theoretical and cryptographic underpinnings of the protocol. With a background in economics and data science, Williams designed Solana's tokenomics model, inflation schedule, and staking reward architecture. He holds advanced degrees and brings an academic rigour to Solana's economic design that is often cited as one of the more thoughtful inflation models in the crypto space. Williams is also a key voice on issues of validator economics, network security, and protocol governance.

Multicoin Capital & Key Contributors

Early Supporters & Core Engineers

Kyle Samani and Tushar Jain of Multicoin Capital were among Solana's earliest and most vocal supporters, investing at the seed round and consistently publishing detailed research defending Solana's technical architecture. Their public advocacy played a key role in building early community and developer credibility. On the protocol side, engineers like Trent Nelson (core validator development), Jack May (Sealevel runtime), and many open-source contributors from across the globe have been essential to building what Solana is today.

04 — THE BLOCKCHAIN TRILEMMA

To understand why Solana's architecture matters, you first need to understand the problem it was designed to solve: the **Blockchain Trilemma**, a term popularised by Ethereum co-founder Vitalik Buterin.

What Is the Trilemma?

The trilemma states that a blockchain system can only ever optimise for **two of the following three properties** simultaneously — never all three at once:

- **Security** — The network is resistant to attacks (51% attacks, double-spends, Sybil attacks). The more validators required to confirm a transaction, the more secure the system.
- **Decentralisation** — No single entity controls the network. Power is distributed among many independent validators, ensuring censorship resistance.
- **Scalability** — The network can handle a high volume of transactions quickly and cheaply. Higher TPS means greater practical utility.

The Trade-offs in Practice

BLOCKCHAIN	SECURITY	DECENTRALISATION	SCALABILITY
Bitcoin	High	High	~7 TPS
Ethereum	High	High	~15-30 TPS
Ripple (XRP)	High	Low (federated)	~1,500 TPS
Solana	High	Debated	65,000+ TPS

Solana's Claim: A Different Approach

Solana does not argue that the trilemma is unsolvable — it argues that the trilemma was framed around the wrong assumption. Most blockchains assumed that **time is an adversarial variable**. Solana's answer: what if you didn't need to trust time from other nodes? What if time itself could be proven, cryptographically, without trusting any single node?

By solving the time problem with Proof of History, Solana eliminates the overhead that forces other blockchains to choose between speed and decentralisation. Validators no longer need to gossip timestamps or wait for agreement on ordering — they can run in parallel and reach consensus faster. The result: high security, competitive decentralisation, and massive scalability — all at once.

Solana's Thesis in One Line

"The blockchain trilemma is not a law of nature — it is a consequence of treating time as an untrusted variable. Solve time, and you solve scalability without sacrificing security or decentralisation."

— Implied by the Solana whitepaper, 2017

05 — HOW SOLANA ACTUALLY WORKS

Solana's performance is not the result of a single innovation — it is the result of **eight interlocking innovations** that work together like the components of a finely engineered machine. Each one solves a specific bottleneck. Together, they achieve what no other production blockchain has matched.

5.1 — Proof of History (PoH)

Proof of History is Solana's most fundamental and original innovation — the one that makes everything else possible. It is not a consensus mechanism. It is a **cryptographic clock** — a way of proving, mathematically, that a specific amount of time has passed between two events, without needing any external party to verify it.

How Does It Work?

PoH works by running a **Verifiable Delay Function (VDF)** — specifically a recursive SHA-256 hash function:

- Start with any initial input (e.g., a random value or the hash of the previous block).
- Compute SHA-256(input) to get Output 1. This takes a fixed, measurable amount of time on any hardware.
- Feed Output 1 back as the next input: SHA-256(Output 1) = Output 2.
- Continue this process indefinitely — each hash is sequentially dependent on the one before it.
- Periodically, record the current hash value and a counter (the 'slot number') creating an immutable verifiable timeline.
- Transaction data can be inserted into the hash chain at any point, stamping that transaction with a verifiable position in time.

Why Is This Revolutionary?

In Bitcoin and Ethereum, validators must communicate with each other to agree on the order and time of transactions. With PoH, the sequence of events is **already proven before consensus even begins**. Validators receive a block and can immediately verify the ordering without waiting for other nodes to confirm timing. This transforms consensus from a slow gossip-based negotiation into a fast parallel verification process.

Analogy: The Atomic Clock

Imagine every country needed to agree on what time it is before any international transaction could settle. Under the old blockchain model, that is essentially what happened.

Proof of History is like giving every validator on the network access to the same atomic clock — one whose readings cannot be faked, because each reading is mathematically derived from the previous one.

You no longer need to argue about what time it is. The math proves it.

5.2 — Proof of Stake & Tower BFT Consensus

Once transactions are ordered by the PoH clock, they still need to be **confirmed** by the network. This is where Solana's consensus mechanism comes in: **Tower BFT** (Byzantine Fault Tolerant), Solana's PoH-optimised adaptation of the classic PBFT consensus algorithm.

Proof of Stake Foundation

Like Ethereum post-Merge, Solana uses Proof of Stake as the economic foundation. Validators must **stake SOL tokens** as collateral to participate in block production and voting. Malicious behaviour results in **slashing** — the destruction of part or all of the validator's staked SOL.

Tower BFT: How Voting Works

- **Lockouts:** When a validator votes for a block, it is locked out from voting for a competing fork for a progressively longer timeout — doubling each time it votes on the same fork.
- **Exponential Lockouts:** After 32 consecutive votes on the same fork, a validator is locked out for 2^32 slots (~136 years), making fork-switching economically irrational.
- **Supermajority:** A block achieves finality when 2/3 of all staked SOL has voted for it — irreversible under normal conditions.
- **PoH Integration:** Because PoH already establishes ordering, validators only need to vote on fork selection, not timing — dramatically reducing message complexity.

5.3 — Turbine: Block Propagation

Once a block is produced by the Leader, it needs to reach every validator as quickly as possible. Solana solves this with **Turbine**, its block propagation protocol inspired by BitTorrent:

- The block is broken into small **shreds** (~1KB fragments each).
- The Leader sends different shreds to different validator **neighbourhoods** (randomly assigned groups).
- Each validator forwards its shreds to a sub-neighbourhood, creating a propagation tree.
- Data fans out exponentially — each validator only needs to send/receive a fraction of total block data.
- Reed-Solomon erasure coding allows full block reconstruction even if some shreds are lost.

5.4 — Gulf Stream: Mempool-less Transaction Forwarding

Solana eliminates the mempool entirely with **Gulf Stream**. Because Solana uses a pre-known **Leader Schedule**, clients and validators can push transactions directly to the expected next Leader — before those transactions are even needed.

- Clients submit transactions to the current and upcoming Leaders rather than broadcasting to the whole network.

- Validators cache forwarded transactions and begin processing them the instant they become Leader.
- Unconfirmed transactions are automatically purged after a few seconds — no mempool bloat.
- Enables Solana to handle **100,000+ pending transactions** without the bottlenecks that plague Ethereum at peak loads.

5.5 — Sealevel: Parallel Smart Contract Execution

Ethereum's Virtual Machine processes transactions **one at a time**, sequentially — a massive inefficiency. Solana's **Sealevel** runtime allows thousands of smart contracts to run **in parallel** across all available CPU and GPU cores simultaneously:

- Every Solana transaction must declare upfront which **accounts** it will read from or write to.
- The Sealevel scheduler identifies transactions that access different accounts (non-overlapping state).
- Non-overlapping transactions are executed simultaneously on different CPU/GPU threads — true parallelism.
- Transactions that access the same account are sequenced to prevent race conditions.
- Scales almost linearly with hardware — more CPU cores means more transactions per second automatically.

5.6 — Pipelining: Transaction Processing Unit

Borrowed from CPU design, **Pipelining** overlaps different stages of transaction processing. Rather than waiting for one transaction to fully complete before beginning the next, Solana uses a four-stage pipeline:

- **Data Fetching** — Incoming transaction data is fetched from the network.
- **Signature Verification** — Cryptographic signatures are verified (GPU-accelerated via Akridge's Seagull system).
- **Banking** — Transactions are processed against account balances and state.
- **Writing** — Updated state is written to disk.

While stage 1 is being processed for Block N, stages 2-4 simultaneously process earlier blocks — no hardware resource sits idle.

5.7 — Cloudbreak: Horizontal Account Database

Solana uses **Cloudbreak**, a custom memory-mapped account database designed for concurrent reads and writes:

- Account data is stored in memory-mapped files — mapped directly into RAM for ultra-fast access.
- Supports concurrent reads and writes across all CPU cores simultaneously.

- Uses a custom data structure optimised for Solana's access patterns (frequent small reads, occasional writes).
- Scales to hundreds of millions of accounts without I/O bottlenecks slowing transaction processing.

5.8 — Archivers: Distributed Ledger Storage

Storing the complete history of a high-TPS blockchain is an enormous challenge. **Archivers** solve this by distributing storage across the network:

- Archiver nodes store subsets of the ledger history — not the full chain.
- They use a Proof-of-Replication system to prove they are storing specific data shards.
- Together, archivers form a distributed, redundant storage layer for the full historical ledger.
- Ensures historical data is always available for audits and indexing without requiring validators to store petabytes of data.

5.9 — The Full Transaction Lifecycle

Here is how all eight innovations work together when a single transaction is submitted — from clicking 'Send' to achieving finality:

1	**User Submits Transaction** A user signs a transaction with their private key and broadcasts it to a Solana RPC node. The transaction declares which accounts it will read/write (required by Sealevel).
2	**Gulf Stream Forwarding** The RPC node uses the published Leader Schedule to identify upcoming Leaders and forwards the signed transaction directly to their TPU ports.
3	**Signature Verification** The Leader's validator hardware queues the transaction for GPU-accelerated signature verification (Akridge's Seagull). Thousands of signatures are verified in parallel.
4	**Banking Stage (Sealevel)** Verified transactions enter the Sealevel scheduler. Non-conflicting transactions are dispatched to parallel CPU threads. Account state is read from Cloudbreak. Transactions execute simultaneously.
5	**Block Assembly & PoH Stamping** The Leader assembles completed transactions into a block and records their results into the PoH hash chain — stamping each with a verifiable sequence number and time.

6	**Turbine Propagation** The block is shredded into ~1KB pieces and propagated across the validator network via Turbine's tree structure. Each validator receives shreds and forwards them onward.
7	**Tower BFT Voting** Validators reconstruct the block from shreds, verify the PoH sequence, re-execute transactions, and cast their Tower BFT vote for this fork.
8	**Optimistic Confirmation (~400ms)** Once ~31% of staked SOL has voted, the transaction achieves optimistic confirmation — extremely unlikely to be reversed. This typically happens within one slot (~400ms).
9	**Full Finality (2-3 seconds)** Once 2/3 of all staked SOL has voted (supermajority), the block achieves full finality. It is now irreversible. The transaction is complete.

06 — VALIDATORS, STAKING & BLOCK PRODUCTION

What Is a Validator?

A **validator** is a computer (or cluster of computers) run by an individual or organisation that participates in Solana's consensus mechanism. Validators are the backbone of the network: they receive transactions, process them, produce blocks, vote on the correct chain, and store ledger history. As of 2025, Solana has over **2,000 active validators** spread across dozens of countries.

Hardware Requirements

Because Solana is a high-performance chain, running a validator requires serious hardware — significantly more powerful than running an Ethereum validator:

COMPONENT	MINIMUM	RECOMMENDED
CPU	12 cores / 24 threads	AMD EPYC / Intel Xeon (24+ cores)
RAM	128 GB	256 GB ECC DDR4
Storage (OS)	500 GB SSD	1 TB NVMe SSD
Storage (Accounts)	2 TB NVMe	2 TB+ NVMe (PCIe 4.0)
Storage (Ledger)	8 TB HDD	16 TB HDD
GPU	Optional	NVIDIA RTX 3090 / A5000 (sig verify)
Network	1 Gbps	10 Gbps symmetric

The Leader Schedule

Solana pre-computes a **Leader Schedule** at the start of each **epoch** (~2 days, 432,000 slots). The schedule assigns each validator a specific set of slots during which it will serve as the block-producing Leader.

- Validators with more staked SOL receive proportionally more Leader slots per epoch.
- The schedule is public and deterministic — all participants know in advance who the next Leader will be.
- This allows Gulf Stream to forward transactions to upcoming Leaders before their turn begins.
- It prevents last-minute manipulation of the Leader selection process.

Staking & Delegation

Not every SOL holder needs to run a validator to earn staking rewards. Solana supports **delegated staking** — where token holders delegate their SOL to a validator of their choice. In return, the delegator receives a share of the validator's block rewards, minus the validator's commission (typically 5–10%).

Rewards & Inflation

- Solana has a disinflationary token emission model. The initial inflation rate was 8% per year at mainnet launch.
- Each year, inflation decreases by 15% of its current rate, targeting a long-term floor of 1.5% annual inflation.
- Block rewards come from newly minted SOL distributed each epoch to validators and their delegators.
- A portion of transaction fees is also distributed to the current block's Leader as additional income.
- 50% of all transaction fees are burned, creating deflationary pressure that counteracts inflation at high usage.

Slashing

To deter malicious behaviour, Solana implements **slashing** — the permanent destruction of a portion of a validator's staked SOL. The primary slashable offences are:

- **Double voting:** Voting for two different blocks in the same slot (attempting to support conflicting forks).
- **Tower BFT violations:** Casting votes that contradict the validator's own earlier lockout commitments.

Slashing has been deliberately kept conservative in Solana's early years to avoid punishing validators for accidental misconfiguration. As the network matures, stricter slashing conditions are expected to be introduced.

This concludes File 1 of the Solana Deep-Dive Research Series — prepared by Nasir Bello.
File 2 covers the SOL token, ecosystem, dApps, and major milestones.
File 3 covers controversies, competitive landscape, and the future outlook.